

Physics Playlist:

Ch 1: sig figs –

- 1.1 Accuracy Precision : https://www.youtube.com/watch?v=PMv69Gne_Bw
- 1.2 Sig Figs: https://www.youtube.com/watch?v=LEBSq_oC9Jk
- 1.3a - Unit Conversions - <https://www.youtube.com/watch?v=lbEn0G-7S4M>
- 1.3b – Units – scale - <https://www.youtube.com/watch?v=-aYat9357mE>
- SI Units song 😊 - <https://www.youtube.com/watch?v=Fcl4gcGN6kk>

Ch 2: 1-D (Linear) Motion–

- 2.1a- speed vs velocity 2 <https://www.youtube.com/watch?v=NZvMXclztuU>
- 2.2a - velocity accel <https://www.youtube.com/watch?v=4dCrkp8qqLU>
- 2.2b - 1D graphs <https://www.youtube.com/watch?v=bqf8m7xNvLg>
- 2.2b - graphs also <https://www.youtube.com/watch?v=VJefeYJL3uE>
- 2.2c - 1D horiz motion- <https://www.youtube.com/watch?v=hpWuZh6oTew>
- A09 - 1D Crash C - <https://www.youtube.com/watch?v=ZM8ECpBuQYE>
- 2.3 Free fall - <https://www.youtube.com/watch?v=E43-CfukEgs> - same room as Avengers where tesseract cube stolen by Loki - <https://www.youtube.com/watch?v=pQKYN-yR2oM>
- 2.3a Felix bung. - <https://www.youtube.com/watch?v=mJxsj51d-Pk>
- 2.3b Wingsuit - https://www.youtube.com/watch?v=u5_ITt2LM0A

Ch 3: 2-Dimensional Motion (Projectile) -

- 3.1 – Review of trig SOHCAHTOA – - <https://www.youtube.com/watch?v=bSM7RNSbWhM>
- 3.1 – solving right triangles - - <https://www.youtube.com/watch?v=a5WQlcFTXyk>
- 3.31 – Horizontal projectile - https://www.youtube.com/watch?v=uEnUG_1TYxc
- 3.32 – practice problem horizontal - <https://www.youtube.com/watch?v=wD2qITZ6w8A&list=PL-mV9m5bis8FRSNBORzz-PzoAmxXJDUO9&index=3>
- 3.33 – Projectile at angle - <https://www.youtube.com/watch?v=aY8z2qO44WA>
- 3.34 – practice problem angle - <https://www.youtube.com/watch?v=TQSvfrSJucI&list=PL-mV9m5bis8FRSNBORzz-PzoAmxXJDUO9&index=1>
- 3.35 – more practice problems - <https://www.youtube.com/watch?v=IQbXXmyf2xc>
- 3.3a Harvard – Shoot the monkey - <https://www.youtube.com/watch?v=OjGZnMf3rPo>
- 3.3b MIT – Shoot the monkey - <https://www.youtube.com/watch?v=cxvsHNRXLjw>
- 3.3c – Projectile – Vx - <https://www.youtube.com/watch?v=KacTRPL1MtE>
- 3.4a – Vector addition Japanese car - <https://www.youtube.com/watch?v=yPHoUbCNPX8>
- 3.4b – Relative motion (truck) - <https://www.youtube.com/watch?v=j1URC2G2qnc>
- 3.4c – bike flip on moving train - https://www.youtube.com/watch?v=TP_0Vv5F29I
- 3.4d – Relative motion ref frame - <https://www.youtube.com/watch?v=wD7C4V9smG4>

3.5a – Relativity #1 – <https://www.youtube.com/watch?v=rBmYUEnafok>

3.5b – Relativity #2 – <https://www.youtube.com/watch?v=iEeSiT3SI4>

3.5c – Relativity #3 – <https://www.youtube.com/watch?v=FPzGaksFCbs>

3.5d – Relativity #4 – <https://www.youtube.com/watch?v=pBQjsOaRHxg>

Lab – micrometer measurement - <https://www.youtube.com/watch?v=Vzze3z1oMSc>

Ch 4:

4.1 Inertia. - <https://www.youtube.com/watch?v=j1URC2G2qnc>

4.1 Newton's 3 laws of motion - <https://www.youtube.com/watch?v=PNTLDNxTMDg>

4.2 Free Body Diagrams - <https://www.youtube.com/watch?v=gDk2te4nrlo>

4.2 FBD – another example - <https://www.youtube.com/watch?v=nDis6HbXxig>

4.4 Friction and coefficient –

Ex 1 - Horizontal friction - <https://www.youtube.com/watch?v=WldiRFIhgYU>

Ex 2 – Horizontal static/kinetic - <https://www.youtube.com/watch?v=RIBeeW1DSZg>

Ex 3 – Horizontal again - <https://www.youtube.com/watch?v=ERifBN-zePQ>

Ex 4 – another horizontal- <https://www.youtube.com/watch?v=WnYsFaXMJn4>

Ex 5 – incline plane – book at rest - <https://www.youtube.com/watch?v=KgOqTB1MIKk>

Ex 5a – drawing diagrams... <https://www.youtube.com/watch?v=52R61aSWHg0>

Ex 6 – another incline plane – <https://www.youtube.com/watch?v=ufgY237M5KQ>

**Ex 7 – yet another... - https://www.youtube.com/watch?v=I8k_lqUiML4

Ex 8 - another 2 <https://www.youtube.com/watch?v=HsSweiqnYIY>

Ex 9 – another 3 - <https://www.youtube.com/watch?v=C0DGuwlIzgc>

**Ex 10 another 4 - <https://www.youtube.com/watch?v=EPORI4WKJPE> **

**Video 10, final answer = $a = 1.45 \text{ m/s}^2$ at 10:45 in video time

Ex 11 – Pulleys - <https://www.youtube.com/watch?v=-Sq870bCnk8>

Ex 12 ... another pulley

<https://www.youtube.com/watch?v=HEqY77RHnV0&list=PLybg94GvOJ9HEEz7qAKsnyREbdtZQGQP9&index=6>

Ex 21 - <https://www.youtube.com/watch?v=8xOU25PWx8M&list=PLX2gX-ftPVXVIUphWULrI1mrL4FB9GzFt&index=2>

**Ex 22 - <https://www.youtube.com/watch?v=7w3RCHHrIII&list=PLX2gX-ftPVXVIUphWULrI1mrL4FB9GzFt&index=3>

Ex 23 - <https://www.youtube.com/watch?v=o-res6KIU9E>

Ex 24 - <https://www.youtube.com/watch?v=WI00eINgtzM>

Ex 25 - <https://www.youtube.com/watch?v=uY0ZkCEL8ZU>

**Ex 26 (minutes 1:07-6 only...) - https://www.youtube.com/watch?v=NddW_MnFhnQ

Ex 27 - <https://www.youtube.com/watch?v=tl3ijqnnxoY>

Forces – concrete - <https://www.youtube.com/watch?v=cZINeaDjisY>

Roads – asphalt - <https://www.youtube.com/watch?v=XKFAC5RYbEM>

Screwdriver – phillips - <https://www.youtube.com/watch?v=R-mDgKtivul>

Potholes - <https://www.youtube.com/watch?v=gRuarpWsKHY>

Ch 5 KE PE

5.1 – Conservation- https://www.youtube.com/watch?v=Zw1bp1V_RwA

5.1a – KE - Penny from Empire State Building -

https://www.youtube.com/watch?v=16Ci_2bN_zc

5.3 Springs – shotgun mechanism - <https://www.youtube.com/watch?v=21uh28Z77Xg>

5.3a – Spring mechanism – cool! <https://www.youtube.com/watch?v=-QTkPfq7w1A>

Ch 6 momentum

6.2 Golf ball #1 – <https://www.youtube.com/watch?v=ZHPK5vvtnf4>

6.2 Golf ball #2 - <https://www.youtube.com/watch?v=c-0dDKfw4NI>

6.2 Golf ball #3 - <https://www.youtube.com/watch?v=JT0wx27J9xs>

Chapter 7 – Circular Motion

7.1 - Centripetal force - https://www.youtube.com/watch?v=zHpAifN_2Sw

7.1a – Roller coaster loops - <https://www.youtube.com/watch?v=4q2W5SJc5j4>

7.2 – Universal Gravitation – <https://www.youtube.com/watch?v=kxkFaBG6a-A>

7.2a – Space and Gravity - <https://www.youtube.com/watch?v=iQOHRKKNNLQ>

7.2b – Earth Spin - <https://www.youtube.com/watch?v=9zso7ChaQXQ>

7.2c – Space Toilet 😊 - <https://www.youtube.com/watch?v=m1wwzwvfsC0>

7.2d - Apollo Rocket - <https://www.youtube.com/watch?v=1nLHIM2IPRY>

7.3 Keplers 3 Laws - <https://www.youtube.com/watch?v=pdst6HQkdrc>

7.3a - Torque - https://www.youtube.com/watch?v=5Zrphnd_OVI

7.3b – Railroad train wheels - <https://www.youtube.com/watch?v=Nteyw40i9So>

7.3c - Angular Momentum - https://www.youtube.com/watch?v=XPUuF_dECVI

7.3e – Soyuz rocket - <https://www.youtube.com/watch?v=v7YgDum2Sg>

7.3f – SpaceX Dragon - https://www.youtube.com/watch?v=Osq_a30_Jd8

7.5 Solar System formation - <https://www.youtube.com/watch?v=gxKCDjnWabk>

7.6 - Torque problems - https://www.youtube.com/watch?v=qGvFAI5CK_c

7.7 Gravitation constant – Cavendish - <https://www.youtube.com/watch?v=70-GBymrck>

7.8 Football spin Veritasium Tom Brady - <https://www.youtube.com/watch?v=J3i3F2e4IYs>

Ch 8/9 Fluids - Heat

- 8.1a Archimedes principle – 1 - <https://www.youtube.com/watch?v=ijj58xD5fDI>
- 8.1b Archimedes principle – 2 - <https://www.youtube.com/watch?v=0v86Yk14rf8>
- 8.1c - Lawn Chair He balloons - https://www.youtube.com/watch?v=2GLrr_Xp0qc

- 8.2a height and pressure - <https://www.youtube.com/watch?v=EBXlrSQIJKs>
- 8.2b Water tower - <https://www.youtube.com/watch?v=yZwfcMSDBHs>
- 8.3a Bernoulli demos - <https://www.youtube.com/watch?v=eKEorBipbO8>
- 8.3b Bernoulli equation - <https://www.youtube.com/watch?v=ytCuHh5PwwY&t=1s>
- 8.3c Continuity - <https://www.youtube.com/watch?v=zgEOiNYw0JA>
- 8.3d Pump - <https://www.youtube.com/watch?v=Vhc-hEjh12I>
- 8.3e gas pump cutoff - <https://www.youtube.com/watch?v=fT2KhJ8W-Kg>
- 8.3f – fluid drag friction - <https://www.youtube.com/watch?v=GMmNKUIXXDs>
- 8.3g – p traps - https://www.youtube.com/watch?v=wlyK1zel_Bk
- 8.3h – base plates - <https://www.youtube.com/watch?v=nGa1244hK9Y>

Misc fluids vids:

- Lift... https://www.youtube.com/watch?v=E3i_XHIVCeU
- Water tower - <https://www.youtube.com/watch?v=yZwfcMSDBHs>
- Continuity - <https://www.youtube.com/watch?v=zgEOiNYw0JA>
- Airlock - <https://www.youtube.com/watch?v=zdkp9N3qfkl>

Physics engine – fluids, power, heat

- Mini engine - <https://www.youtube.com/watch?v=guglj7TAZ2o>
- carburetor - <https://www.youtube.com/watch?v=toVfvRhWbj8>
- fuel injection - <https://www.youtube.com/watch?v=N0RIGWUnVFc>
- hydroelectric model - <https://www.youtube.com/watch?v=Z2xIO2kB2lw>
- Excavator - Pressure F/A = https://www.youtube.com/watch?v=3Eo_GvDGP-g
- Heat – expansion - <https://www.youtube.com/watch?v=pH7VfJDq7f4>
- Solar power - https://www.youtube.com/watch?v=ccO0_hSXMMyM

Waves –

- 11.1 - Pendulum – cool - <https://www.youtube.com/watch?v=yVkdFJ9PkRQ>
- 11.1a – another pendulum - https://www.youtube.com/watch?v=4ICFmAr_DQ
- 11.3 interference - <https://www.youtube.com/watch?v=eW5VGGJuWtQ>
- 11.3a nodes - ** <https://www.youtube.com/watch?v=0Rfushlee0U>

- Cell phone waves - <https://www.youtube.com/watch?v=0faCad2kKeg>

Sound-

12.1 – doppler1 - <https://www.youtube.com/watch?v=h4OnBYrbCjY>

12.1a – doppler 2 - <https://www.youtube.com/watch?v=7l1OxhxCh1E>

12.1b – doppler 3 - <https://www.youtube.com/watch?v=fWX38qjECyE>

12.1c - doppler 4 - https://www.youtube.com/watch?v=SqGtp6RQy_8

12.2 – standing waves - <https://www.youtube.com/watch?v=0Rfushlee0U>

12.2a – slinky standing waves - <https://www.youtube.com/watch?v=-k2TuJfNQ9s>

12.2b – decibels - <https://www.youtube.com/watch?v=WZLQoP6CM0k>

12.2c Resonance – Tacoma Narrows Bridge -

<https://www.youtube.com/watch?v=mXTSnZgrfxM>

12.2d Resonance 1 - https://www.youtube.com/watch?v=B_u3sGbpM8M

12.2e Resonance 2 myth busters - <https://www.youtube.com/watch?v=fr3gWpxuJxE>

Sound harmonics

12.3 – beats - <https://www.youtube.com/watch?v=4M72kQulGKk>

12.3a – guitar tuning beats --- <https://www.youtube.com/watch?v=rmvDu6EY2IE>

12.3d Speed of sound – whip it good - https://www.youtube.com/watch?v=AnaASTBn_K4

12.3e 🍡 Sugar Plum Fairy - https://www.youtube.com/watch?v=QdoTdG_VNV4

12.3f Glass armonica - <https://www.youtube.com/watch?v=eEKIRUvk9zc>

12.3g Singing plates - <https://www.youtube.com/watch?v=wYoxOJDrZzw>

12.3h cicada sound - <https://www.youtube.com/watch?v=TWc48iVC8u8>

13: Mirrors

13.1 Mirror

13.4 Color polarization:

13.4a Polarized light 1 - <https://www.youtube.com/watch?v=c9ew1J0PY-M>

13.4b Polarized light 2 - <https://www.youtube.com/watch?v=CSu0cV3fqj8>

Why is the sky blue? -

Part 1 - <https://www.youtube.com/watch?v=4HBuHX4-VU8>

Part 2 - <https://www.youtube.com/watch?v=UvmWxm3nR6E>

Light-

14.1 Speed of light - <https://www.youtube.com/watch?v=pTn6Ewhb27k>

14.1a speed part 2 <https://www.youtube.com/watch?v=jUHgIYNEzJQ>

14.1b -speed part 3. <https://www.youtube.com/watch?v=kZZ4B8UiEGM>

14.3c Microwave Oven: - <https://www.youtube.com/watch?v=kp33ZprO0Ck>

14.4 Grape Microwave-total internal reflection –

<https://www.youtube.com/watch?v=wCrtk-pyP0I>

14.3s – total internal reflection aquarium - <https://www.youtube.com/watch?v=RhRI0gxGOAA>

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Mirrors lenses

1. <https://www.youtube.com/watch?v=AMrhQ5d2N48>
2. <https://www.youtube.com/watch?v=7GV1UZSTNJg>
3. https://www.youtube.com/watch?v=GPomGcZ3_kk
4. Showing images - <https://www.youtube.com/watch?v=fyYaDeTonVQ>

Circuits –

<https://www.youtube.com/watch?v=kjW4H3fKi8o>

<https://www.youtube.com/watch?v=-PiB2Xd3P94>

Electricity – Power lines - <https://www.youtube.com/watch?v=bHlhgxav9LY>

Cell phone waves - <https://www.youtube.com/watch?v=0faCad2kKeg>

Fiber optics – a - https://www.youtube.com/watch?v=0MwMkBET_5I

Misc:

Electricity – storage - <https://www.youtube.com/watch?v=66YRCjxlcg>

GPS – relativity - <https://www.youtube.com/watch?v=ozAPGnr-934>

Turbine – 3blade - <https://www.youtube.com/watch?v=RNPIRfxUTQ4>

Turbine- assembly - <https://www.youtube.com/watch?v=0vE6QkvcV-s>

DC motor - <https://www.youtube.com/watch?v=CWulQ1ZSE3c>

Plywood produced - <https://www.youtube.com/watch?v=eF5LVBW1vI8>

Plywood again - <https://www.youtube.com/watch?v=3Wh9NYvfStk>

Misc:

Tattoo slow mo – <https://www.youtube.com/watch?v=kxLoycj4pJY>

Tattoo removal - <https://www.youtube.com/watch?v=D0B7F5UbTOQ>

Microscope - skin, etc.- <https://www.youtube.com/watch?v=DfSnHylryeY>

Marvel Universe and Science - <https://www.youtube.com/watch?v=FGP4Klkr2Is>